

○ Why Python?

- Python works on different platforms (Windows, Mac, Linux, Raspberry Pi, etc).
- Python has a simple syntax similar to the English language.
- Python has syntax that allows developers to write programs with fewer lines than some other programming languages.
- Python runs on an interpreter system, meaning that code can be executed as soon as it is written. This means that prototyping can be very quick.
- Python can be treated in a procedural way, an object-oriented way or a functional way.

Things to know

- The most recent major version of Python is Python 3, which we shall be using .
- However, Python 2, although not being updated with anything other than security updates, is still quite popular.
- Python will be written in a text editor.
- It is possible to write Python in an Integrated Development Environment, such as Thonny, Pycharm, Netbeans or Eclipse which are particularly useful when managing larger collections of Python files.



- Example
- `print("Hello, World!")`
- **Python Install**
- Many PCs and Macs will have python already installed.
- To check if you have python installed on a Windows PC, search in the start bar for Python or run the following on the Command Line (cmd.exe):

```
C:\Users\Your Name>python -version
```

If you find that you do not have Python installed on your computer, then you can download it for free from the following website: <https://www.python.org/>



○ **Verify installation**

- Once IDLE is launched, go to →file menu
- Click on → new file to create new python file.

○ **How to use IDLE**

- Print("morning")
- Save the file with .py extension and run the file(run menu→ click on run module or press F5)
- If the output is morning then installation is successful and IDLE is ready to use.

○ **Python Quickstart**

- Python is an interpreted programming language, this means that as a developer you write Python (.py) files in a text editor and then put those files into the python interpreter to be executed.
- The way to run a python file is like this on the command line:
- C:\Users\panka>AppData\Local\Programs\Python\Python311\my.py
- Where "my.py" is the name of your python file.
- Let's write our first Python file, called my.py, which can be done in any text editor.



- **The Python Command Line**

- To test a short amount of code in python sometimes it is quickest and easiest not to write the code in a file. This is made possible because Python can be run as a command line itself.
- Type the following on the Windows, Mac or Linux command line:
- **C:\Users\Your Name>python**
- Or, if the "python" command did not work, you can try "py":
- **C:\Users\Your Name>py**
- From there you can write any python, including our hello world example from earlier in the tutorial:
- **>>> print("Hello, World!")**
- Which will write "Hello, World!" in the command line:
- **Hello, World!**
- Whenever you are done in the python command line, you can simply type the following to quit the python command line interface:
- **>>>exit()**



- **Features and Benefits of Python**

- Compatible with a variety of platforms including Windows, Mac, Linux, Raspberry Pi, and others
- Uses a simple syntax comparable to the English language that lets developers use fewer lines than other programming languages
- Operates on an interpreter system that allows code to be executed immediately, fast-tracking prototyping
- Can be handled in a procedural, object-orientated, or functional way

- **Python Identifiers**

- Identifiers are the name given to variables, classes, methods, functions etc. For example

```
language = 'Python'
```

- Here, language is a variable (an identifier) which holds the value 'Python'.
- We cannot use keywords as variable names as they are reserved names that are built-in to Python. For example,

```
continue = 'Python'
```

- The above code is wrong because we have used continue as a variable name. (CONTINUE IS KEYWORD)